

**GRAPH THEORY:
HOMEWORK 4
BOOK SECTION 2.1.**

ROB TOMPKINS

1. Prove that every tree with maximum degree $\Delta > 1$ has at least Δ vertices of degree 1. Show that this is the best possible by constructing an n -vertex tree with exactly Δ leaves, for each choice of n, Δ with $n > \Delta \geq 2$.

Proof. Let v be a vertex of degree Δ . Deleting v from the tree leaves exactly Δ components, one containing each neighbor of v : two such neighbors cannot lie in the same component, since a path between them avoiding v would give a cycle. In each component choose a vertex x whose distance from v is largest. The unique v - x path gives x one neighbor toward v . Any other neighbor would be farther from v , contradicting the choice of x . Thus x has degree 1 in the original tree. The components are disjoint, so these give at least Δ leaves.

For sharpness, start with the star $K_{1,\Delta}$, which has $\Delta + 1$ vertices and exactly Δ leaves. Subdivide one edge $n - \Delta - 1$ times. Each subdivision adds one vertex of degree 2 and preserves all existing vertex degrees. The resulting tree has n vertices, maximum degree Δ , and exactly Δ leaves. $\square$

2. Prove that every graph with diameter d has an independent set of size at least $\lceil (1+d)/2 \rceil$.

Proof. Choose vertices at distance d , and let $v_0, v_1, \dots, v_d$ be a shortest path between them. No two nonconsecutive vertices of this path are adjacent: an edge $v_i v_j$ with $j \geq i + 2$ would replace a segment of length $j - i$ by one edge and give a shorter v_0 - v_d path. Consequently,

$$\{v_0, v_2, v_4, \dots, v_{2\lfloor d/2 \rfloor}\}$$

is independent. Its size is $\lfloor d/2 \rfloor + 1 = \lceil (1+d)/2 \rceil$, as required. Here, as usual, the diameter is assumed to be finite. $\square$

3. Let $d_1, \dots, d_n$ be positive integers with $n \geq 2$. Prove that there exists a tree with vertex degrees $d_1, \dots, d_n$ if and only if $\sum d_i = 2n - 2$.

Proof. If a tree has n vertices, then it has $n - 1$ edges. The handshaking lemma therefore gives

$$\sum_{i=1}^n d_i = 2|E| = 2n - 2.$$

Conversely, suppose the positive integers $d_1, \dots, d_n$ sum to $2n - 2$. We use induction on n . When $n = 2$, positivity forces $d_1 = d_2 = 1$, and the single-edge tree has these degrees.

Suppose $n \geq 3$. At least one d_i is 1, since otherwise the sum would be at least $2n$. At least one other entry is at least 2, since $2n - 2 > n$. Relabel so that $d_n = 1$ and $d_1 \geq 2$. The $n - 1$ positive integers

$$d_1 - 1, d_2, \dots, d_{n-1}$$

have sum $2n-4 = 2(n-1)-2$. By induction there is a tree on vertices $v_1, \dots, v_{n-1}$ with these degrees. Add a new vertex v_n and the single edge v_1v_n . Adding a leaf preserves connectedness and creates no cycle, so the result is a tree. Its degrees are precisely $d_1, \dots, d_n$. $\square$

4.

- (a) Prove that every tree is bipartite.
- (b) Prove that every tree T has a leaf in its larger partite set (in both if they have equal size).

Proof. (a) Fix a root v and partition the vertices into sets A and B according to whether their distance from v is even or odd. Every edge of a tree joins a vertex to its predecessor on the unique path from the root, so its endpoints have distances from v differing by 1. Thus every edge joins A to B , proving that the tree is bipartite.

(b) Assume T has at least two vertices, and let (A, B) be its bipartition with $|A| \geq |B|$. All vertices have positive degree. If A contained no leaf, every vertex in A would have degree at least 2. Since each edge has exactly one endpoint in A , this would imply

$$|A| + |B| - 1 = |E(T)| = \sum_{x \in A} \deg(x) \geq 2|A| \geq |A| + |B|,$$

a contradiction. Hence A contains a leaf. If $|A| = |B|$, the same argument with the two sets interchanged shows that B also contains a leaf.

Remark. The assumption of at least two vertices is necessary in (b): the one-vertex tree has no vertex of degree 1. $\square$

5.

- (a) Prove that the distance function $d(u, v)$ on pairs of vertices of a graph satisfies the *triangle inequality*: $d(u, v) + d(v, w) \geq d(u, w)$.
- (b) Use (a) to prove that $\text{diam}(G) \leq 2 \cdot \text{rad}(G)$ for every graph G .
- (c) For all positive integers r and d that satisfy $r \leq d \leq 2r$, construct a simple graph with radius r and diameter d . (Hint: Build a suitable graph with one cycle.)

Proof. (a) Concatenate a shortest u - v path and a shortest v - w path. This gives a u - w walk of length $d(u, v) + d(v, w)$. Deleting any closed portions of the walk leaves a u - w path of no greater length. By the definition of distance,

$$d(u, w) \leq d(u, v) + d(v, w).$$

This argument applies when the distances on the right are finite; if either is infinite, the inequality holds with the usual extended-distance convention.

(b) For a connected graph, write $e(x) = \max_y d(x, y)$ for the eccentricity of x , and choose a vertex c with $e(c) = \text{rad}(G)$. For any vertices u, w , part (a) gives

$$d(u, w) \leq d(u, c) + d(c, w) \leq 2e(c) = 2\text{rad}(G).$$

Taking the maximum over u, w proves $\text{diam}(G) \leq 2\text{rad}(G)$. If disconnected graphs are allowed with infinite distances, both radius and diameter are infinite, and the inequality also holds in that convention.

(c) Set $\ell = d - r$, so $0 \leq \ell \leq r$. Start with the cycle C_{2r+1} and choose a vertex c on it. Attach a path of length ℓ at c , using new vertices $x_1, \dots, x_\ell$ and edges $cx_1, x_1x_2, \dots, x_{\ell-1}x_\ell$.

When $\ell = 0$, attach nothing. Since $r \geq 1$, the cycle has at least three vertices; hence this is a simple graph with exactly one cycle.

Distances between cycle vertices are unchanged by the attached path, and every cycle vertex has a cycle vertex at distance r . Also, for a cycle vertex y and $1 \leq i \leq \ell$,

$$d(x_i, y) = i + d(c, y),$$

because every path from the attached path to the cycle passes through c . Thus every cycle vertex has eccentricity at least r , and every new vertex x_i has eccentricity at least $i + r$. Meanwhile,

$$e(c) = \max\{r, \ell\} = r.$$

Therefore the radius is exactly r .

Two cycle vertices are at distance at most r , two vertices of the attached path (including c) are at distance at most ℓ , and a new vertex and a cycle vertex are at distance at most $\ell + r = d$. If $\ell > 0$, choose a cycle vertex y at distance r from c ; then $d(x_\ell, y) = \ell + r = d$. If $\ell = 0$, the cycle itself has diameter $r = d$. Hence the constructed graph has diameter exactly d . $\square$